

Lab 7 Map visualisation

2026-01-18

Week 7 Visualisation with maps

Following on from the considerations in preprocessing

Can I produce a quick “first map” just to confirm things are wired up?

Can I transform my data into the required format if needed?

What’s the simplest version of this map I can plot now?

You’ll also need basemap tiles or other geographic layers to draw the map. We’ll use OpenStreetMap for its permissive licence.

Always consider licensing and operational constraints:

Licensing & cost: Is there a charge for use or redistribution?

Connectivity: Can it run locally, or does it require an internet connection?

Data handling: Does it send your data to a third party?

Risk: Even for basic projects, assess the risks and trade-offs before you choose a source.

Please note: I would not consider myself an expert in licensing and regulation, and these exercises are not intended to be used as such.

Imports

```
library(plotly)

file_id <- "1PNde4IgobljN9xqDCBQjm74-cGTJ2dbJ"
url <- paste0("https://drive.google.com/uc?id=", file_id)

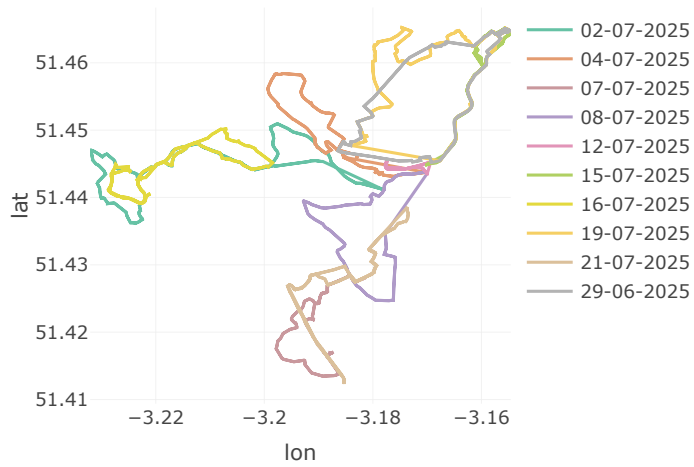
map_df <- read.csv(url)

head(map_df)
```

```
##      lat      lon      date
## 1 51.43859 -3.17363 21-07-2025
## 2 51.43856 -3.17366 21-07-2025
## 3 51.43844 -3.17372 21-07-2025
## 4 51.43835 -3.17374 21-07-2025
## 5 51.43821 -3.17372 21-07-2025
## 6 51.43816 -3.17372 21-07-2025
```

1. Plot the latitude and londitudes

To start to explore plotting geoline information we can simply plot our dataset on a line chart.



2. Add Map Tiles

In Plotly , adding map tiles is straightforward simply add `type = scattermapbox`.

You will also need to specify a mapbox style in layout ie `layout(mapbox = list(style = "open-street-map"))`. In this example, we use 'open-street-map', which is free, open-source, and easy to use. This style provides the background map tiles for the visualisation.

Examples of different map styles and map-based plots can be found here:

<https://plotly.com/r/scattermapbox/>

Recommendation:

Maps can become complex very quickly. Start with a simple map and gradually add features as your understanding develops.

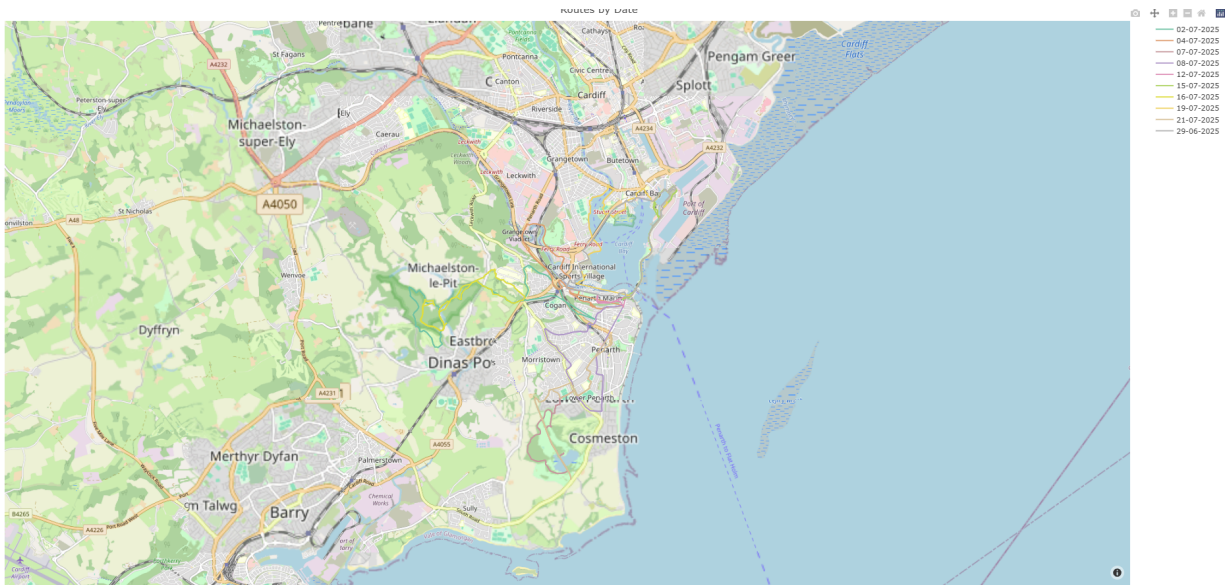


Figure 1: map